

Yifan Chang

6 Years of Front-End Development | Deep Vue 2 / 3 Expertise | Complex B-End & Data Visualization

📍 Hangzhou 📞 152-3798-2380 ✉ 15237982380@163.com

Career Objective

Target Position: Senior Front-End Engineer **Preferred City:** Hangzhou **Expected Salary:** Negotiable

Availability: Within 1 month

Profile

Front-end engineer with 6 years of experience and deep expertise across the Vue 2 / 3 ecosystem. Have led front-end delivery for a range of large-scale B/C-end projects spanning medical pathology, laboratory data platforms, government-procurement big data, and brand e-commerce. Proficient across the full chain of architecture design, engineering, and performance optimization, with hands-on delivery experience in large-file chunked upload, SSR, Web Worker / SharedWorker, ECharts large-screen dashboards, and DICOM medical imaging. Able to independently own complex modules end-to-end, from solution design through production release.

Technical Skills

- **Core Frameworks:** Proficient in Vue 2.x / Vue 3.x (Composition API, Pinia / Vuex, Vue Router), with an in-depth understanding of the reactivity system, virtual DOM diffing, and the scheduling mechanism; familiar with the React 18 Hooks ecosystem.
- **Language Foundation:** Skilled in JavaScript (ES6+) and TypeScript; solid HTML5 / CSS3 (Flex / Grid / animation / Sass / Less / UnoCSS).
- **Engineering:** Build optimization and from-scratch setup with Webpack 2 / 3 / 5 and Vite; Babel, ESLint, Prettier, and Husky workflows; Monorepo (pnpm workspace).
- **Multi-Platform:** Cross-platform development with UNI-APP (WeChat Mini Program / Baidu Mini Program / H5) and Electron; Vue SSR isomorphic rendering (vue-server-renderer + Express).
- **UI / Visualization:** Element Plus / Ant Design Vue / iView / Vant; ECharts (including map drill-down and Rich Text annotations), v-charts, AntV, heatmap.js.
- **Performance & Multithreading:** Web Worker / SharedWorker / Service Worker; SparkMD5 chunk verification; RxJS applied to the request layer, error streams, and duplicate-submission prevention; virtual scrolling.
- **Network & State:** Deep encapsulation of Axios interceptors (authentication, token renewal, CancelToken race-condition prevention); STOMP WebSocket; Promise-based IndexedDB ORM; multi-level local caching strategies.
- **Vertical Expertise:** Cornerstone.js suite (core / tools / wado-image-loader) for DICOM medical imaging; the KFB digital-slide format; ABP Framework multi-tenant enterprise system integration.
- **Collaboration Tools:** Git / GitLab CI, Jenkins, and Docker fundamentals; Swagger integration testing, Postman, and Charles packet capture.

Work Experience

Company: Information-based SaaS solutions for the medical-pathology / clinical-testing industry; flagship products include the PIMS pathology information management system and a laboratory data management platform.

Project 1: PIMS Pathology Information Management System (Vue 3 + Webpack 5 + Element Plus + Vuex 4 + UnoCSS)

- ▶ **Background:** A full-process information platform for hospital pathology departments, covering 30+ business modules including registration, sampling, slide preparation, diagnosis, reporting, archiving, and quality control; supports multiple hospital campuses, roles, and themes. Codebase of 1,455+ files, serving multiple Class-A tertiary hospitals.
- ▶ Led the front-end architecture and engineering system: Webpack 5 splitChunks on-demand code splitting + UnoCSS atomic CSS + git-revision version injection, reducing first-screen JS size by ~35%.
- ▶ Designed a four-layer dynamic routing permission system (whitelist / API-driven menus / role differentiation / independent business modules), with forkJoin parallel initialization + runtime router.addRoute injection, covering dual page-level + button-level authorization across 30+ modules.
- ▶ Introduced SharedWorker for multi-tab login-state synchronization; built a custom Promise-based IndexedDB ORM (20+ ObjectStores) for layered local caching of search history, report drafts, and tile-board cache.
- ▶ Precisely applied RxJS in HTTP interception, duplicate-submission prevention, and WebSocket state transitions (Subject + throttleTime for error merging, Map + setTimeout idempotency lock, distinctUntilChanged).

Project 2: Laboratory Data Management Platform (Vue 2.7 + TypeScript + Vuex + ABP Framework + Webpack)

- ▶ **Background:** A multi-tenant, enterprise-grade laboratory data management platform integrated with an ABP Framework backend, supporting both domain-account SSO and platform-account login, serving multiple corporate clients.
- ▶ Built a closed-loop dual-mode login (SSO + platform account): exchanging a URL code for a token + AES-encrypted password login to ABP; recording consumed OAuth codes in localStorage to prevent idempotency issues; the entire flow encapsulated in a Vuex action chain, transparent to callers.
- ▶ Deep ABP Framework integration: fetching global configuration (permissions / localization / multi-tenancy) on startup and deep-merging it into the abp object before mounting Vue; Axios automatically injects TenantId / Culture headers, making multi-tenancy transparent to business code.
- ▶ Built a multi-environment + multi-variant build system (7 .env sets + entry- and component-level conditional replacement); replaced contenthash with a build timestamp to resolve release-update issues under CDN strong caching, so users automatically load the latest version.
- ▶ Encapsulated a global utility library via the Webpack ProvidePlugin (with matching TS declarations); defined a tiered selection strategy of iView Table / VXE-Table (virtual scrolling) for large-data scenarios.

Company: End-to-end e-commerce transaction system solutions for international brands such as Dell / Alienware, covering mini programs, H5, and PC back-office.

Project 3: Dell / Alienware Brand E-Commerce Mini Program (Vue 2 + UNI-APP + Vuex; isomorphic across WeChat / Baidu / H5)

- ▶ **Background:** A dual-brand (Dell and Alienware) e-commerce mini program covering the full transaction chain

(products, SKU, cart, orders, returns/exchanges, coupons, VIP membership, live streaming, etc.), with 100+ API endpoints and 10+ business modules.

- ▶ Built a “multi-brand × multi-platform” single-codebase publishing architecture: UNI-APP conditional compilation produces 2 brands × 3 platforms = 6 build outputs from one repository; encapsulated a build.js multi-brand packaging script handling AppID switching → compilation → output in a single command.
- ▶ Encapsulated a unified request middleware: dynamic API BaseURL switching eliminates hard-coded mis-release risk; silent re-login on a 401 Token + an autoLoginLoading request lock resolves concurrent-expiry scenarios; 12 fields automatically injected into request headers.
- ▶ Governed 4 analytics-tracking SDKs (Sensors / EUB / JiNGsocial / SR) with centralized post-login initialization and identity synchronization; built a custom observeExpOnce based on IntersectionObserver for precise product-card exposure tracking.
- ▶ Mini-program-specific capabilities: tiered login-state management for the VIP membership system; unified management of nearly a hundred subscription-message tmplId mappings; Canvas poster generation (roundRect / drawText / remote-image base64); cross-mini-program navigation + H5 fallback adaptation.

Project 4: Fancy BOS E-Commerce Back-Office Management System *(Vue 2 + hand-built Webpack 3 + Element UI + ECharts)*

- ▶ **Background:** Covers the full e-commerce chain — PMS (products) / OMS (orders) / SMS (marketing) / CMS (content) / UMS (permissions) / membership / live streaming / data analytics — with a routing table of nearly 2,300 lines; a large, multi-module back-office system.
- ▶ OSS client-side direct-upload solution: the front end POSTs directly to the OSS bucket using a temporary Policy issued by the backend, with zero server traffic involved; also supports a MinIO private-storage mode, switching between two upload paths via a useOss flag.
- ▶ Accumulated a reusable component library of 10+ highly reused components across the business: multiUpload / draggableUpload / TinyMce / excelExport (lazy-loaded) / choose-goods / statisticDate, etc., significantly improving cross-module reuse.
- ▶ Global Axios request cancellation: CancelToken registered to window._axiosPromiseArr to uniformly cancel on route change, avoiding cross-page async races; responses detected as octet-stream are auto-passed through for caller-side download.
- ▶ Hand-built on Webpack 3 (no Vue CLI dependency), maintaining four independent build pipelines for dev / test / pre / prod; integrated v-charts + heatmap.js for multi-dimensional dashboards and user-behavior heatmaps.

Fujian Boss Software Co., Ltd. | Front-End Engineer

2020.09 — 2022.01

Company: *Big-data analytics and visualization solutions for the government-procurement sector.*

Project 5: Government-Procurement Big-Data Analytics Platform (Web + Mobile) *(Vue 2.6 + Vue SSR + Webpack 2 + Express + ECharts)*

- ▶ **Background:** A multi-dimensional visualization analytics platform covering the full government-procurement data lifecycle, supporting both a Web large-screen and mobile, integrated with multi-region, multi-year government data.
- ▶ Helped build Vue 2 SSR isomorphic rendering: a dual-entry Webpack build with vue-server-renderer + Express; window.__INITIAL_STATE__ takes over the server-side Vuex for zero-blank-screen hydration; asyncData + Promise.all parallel prefetch avoids a second client-side request.

- ▶ Two-level server-side caching: component-level LRU (max 1000 / TTL 15 min) + route-level Micro-Cache (1-second granularity); a custom setup-dev-server.js enables hot reload across both SSR compilers simultaneously; three-stage CommonsChunkPlugin splitting + [chunkhash] for long-term caching.
- ▶ Deeply encapsulated an ECharts map component: an extendsMap method supporting province → city → district three-level drill-down; GeoJSON dynamic import for on-demand loading; a bussType prop to switch multi-sector Rich Text annotations — a single component reused across 5+ large-screen pages system-wide.
- ▶ Key mobile capabilities: a responsive solution using PostCSS pxtoem + a 4-breakpoint dynamic rem baseline; manual pinch-to-zoom (Math.hypot + requestAnimationFrame); a vuedraggable dashboard + localStorage layout persistence.

Key Project Highlights

★ GB-Scale Large-File Chunked Upload Based on Web Worker + SparkMD5 — PIMS

- ▶ **Background:** Pathology WSI images often reach GB-scale per file; computing MD5 synchronously on the main thread froze the page for long periods, and traditional uploads had high failure rates.
- ▶ **Solution:** A dedicated Worker (upload.worker.js) uses SparkMD5 to incrementally compute the whole-file fingerprint, with an additional per-chunk MD5 integrity check before each chunk upload; a server-returned “lack” array drives resumable uploads; circuit-breaking after 5 consecutive failures; upload progress is pushed via postMessage and turned into a reactive progress stream with an RxJS Subject.
- ▶ **Impact:** Main-thread FPS stayed at a steady 60 throughout uploads; the GB-scale upload success rate rose from ~65% to over 99%; users can continue other tasks during upload.

★ Imperative Dialog Service Based on Vue 3 render Functions (Headless Dialog) — PIMS

- ▶ **Background:** B-end systems invoke dialogs intensively; the declarative template approach led to fragmented code, low reuse, and difficult context management.
- ▶ **Solution:** hzttDialog uses createApp + render functions to imperatively mount/unmount Vue component trees from JS; manually injecting vNode.appContext = window.hzttApp._context lets dialogs use \$t, global components, and other context; render(null, container) on close actively unmounts to prevent memory leaks; derived higher-order dialogs such as hzttFormDialog / hzttTableDialog / hzttCameraDialog.
- ▶ **Impact:** Unified the invocation pattern for 200+ dialogs project-wide, saving ~30 lines of template code per dialog on average and significantly improving component reuse.

★ Token Sliding-Renewal Axios Encapsulation — Eliminating the “Phantom Logout” False Bug — Laboratory

Data Platform

- ▶ **Background:** Without renewal, the browser silently clears the token after the cookie expires; the next request sends an empty header, the backend returns 401, and the user experiences a hard-to-reproduce “suddenly logged out mid-task” false bug.
- ▶ **Solution:** Encapsulated a unified Axios instance; the request interceptor auto-injects Authorization / Abp.TenantId / .AspNetCore.Culture, hiding multi-tenancy and auth details; the response interceptor implements token sliding renewal — each successful request calls abp.auth.setRefreshToken() to extend cookie validity by 24 hours; error handling is unified in the interceptor: 401 clears state via dialog, 403 shows an insufficient-permission notice, network errors prompt retry — zero branching in the component layer.
- ▶ **Impact:** “Phantom login state” tickets were fully eliminated; login renewal is completely transparent to users;

error-handling code in business components dropped by ~70%.

★ Browser-Side DICOM Medical Imaging Integration (a Scarce B-End Vertical Capability) — PIMS

- ▶ **Background:** Pathology / radiology doctors need to view standard DICOM images and perform professional operations directly in the browser; no off-the-shelf Vue 3 solution exists in the industry.
- ▶ **Solution:** Integrated the full Cornerstone.js ecosystem (core / math / tools / wado-image-loader) for browser-side DICOM rendering; encapsulated a WADO protocol adaptation layer; integrated a viewer for the proprietary KFB digital-slide format; built a custom toolbar component supporting window width/level, zoom, pan, annotation, and other professional operations; combined with STOMP WebSocket for real-time imaging-status push.
- ▶ **Impact:** Covers 90%+ of pathology-department image-review scenarios, sparing doctors from switching to desktop professional software and improving per-case review efficiency by ~50%.

★ Online Page-Lag Performance Investigation — Tracing the Root Cause Across Layers to the CI/CD

Infrastructure — *Laboratory Data Platform*

- ▶ **Background:** Users reported noticeable lag on online pages while the local environment was entirely fine; initial suspicion fell on component rendering or API latency.
- ▶ **Solution:** Recorded a full operation snapshot with the Chrome Performance panel and analyzed the flame graph frame by frame, locating Long Tasks concentrated in JS parsing and script execution rather than rendering or the network; further compared local and online build artifacts and found an abnormal JS syntax-downgrade level in the online bundle; ultimately traced it to the Jenkins pipeline on the ops side, pinpointing the root cause as an outdated Node.js version used for packaging that prevented some dependencies from completing modern-syntax compilation, outputting inefficient polyfills that slowed the main thread.
- ▶ **Impact:** Upgrading the CI pipeline's Node version completely eliminated the page lag; distilled a layered investigation methodology of "browser layer → artifact-comparison layer → build-environment layer."

Education

Luoyang Institute of Science and Technology Computer Science and Technology Bachelor 2018.06 — 2020.09

Self-Assessment

- ▶ Systematic understanding of the Vue 2 / 3 reactivity system, virtual DOM, and scheduling mechanism, able to locate and solve problems at the framework level; familiar with the real-world implementation of performance and engineering solutions such as Web Worker / SharedWorker / RxJS / IndexedDB.
- ▶ Full-chain capability spanning architecture design, engineering setup, performance optimization, and tough-problem investigation; once independently traced an online lag issue from the front-end code layer down to the CI/CD infrastructure layer.
- ▶ Deep accumulation in vertical-industry projects (medical pathology, government procurement, brand e-commerce), able to independently deliver complete solutions in scenarios such as DICOM imaging, complex business modeling, multi-tenant isolation, and large-screen visualization.
- ▶ Focused on balancing DX (developer experience) and UX (user experience), with a habit of distilling repetitive

work into tools, commands, or components; continuously follows the Vue / Vite / TC39 community.

- ▶ Strong cross-team collaboration and documentation habits, proactively driving the adoption of coding standards, code review, and team tech sharing.